



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
08-11-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

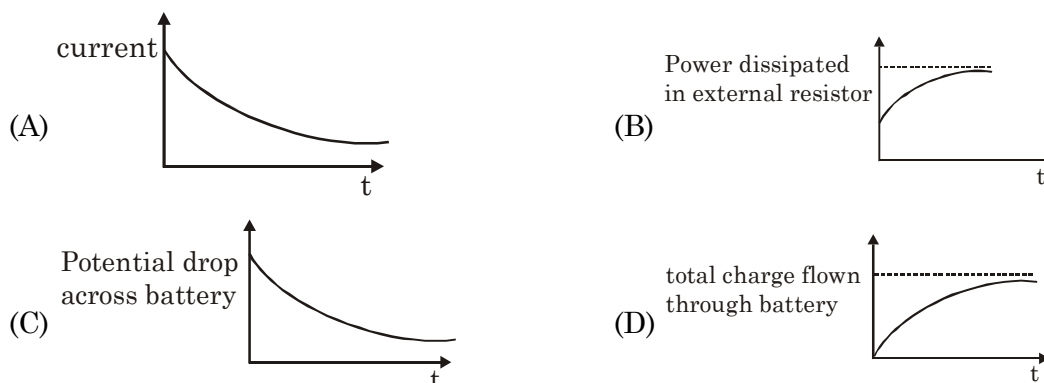
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

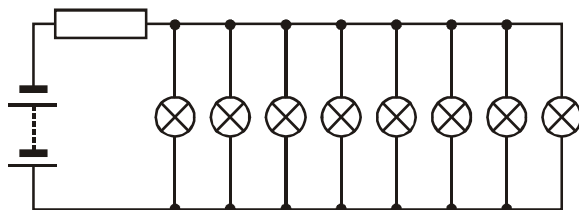
- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The internal resistance of a battery changes with the state of its discharge. If the time for which it is operated is t , the internal resistance is $1.4 + \frac{t}{1000} \Omega$, here t is time in seconds. The emf remains constant at 5V. Which of the following is a correct graph when the battery is connected to a constant resistor of resistance 10Ω .

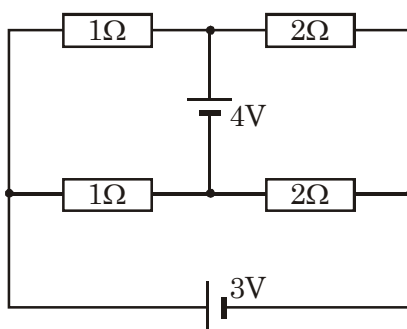


Space for Rough Work

2. Eight identical fluorescent lamps are connected to a constant voltage source as shown in the figure below. There is a resistor in series with the voltage source with resistance half of a single lamp. Ignore the temperature dependence of the resistance of the lamps.



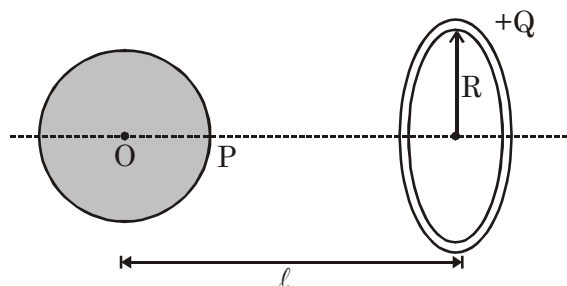
- (A) If 2 lamps burn out, the total power emitted by lamps increase.
 (B) If 6 lamps burn out, the total power emitted by lamps increase.
 (C) If 7 lamps burn out, the total power emitted by lamps decreases in comparison to case in (B)
 (D) If 4 lamps burn out, the total power emitted by lamps increase.
3. Determine the currents through the batteries.



- (A) $i_4 = 3\text{A}$ (Through 4V battery)
 (B) $i_3 = 2\text{A}$ (Through 3V battery)
 (C) The 4V battery is getting discharged and 3V battery is getting charged
 (D) Both batteries are getting discharged

Space for Rough Work

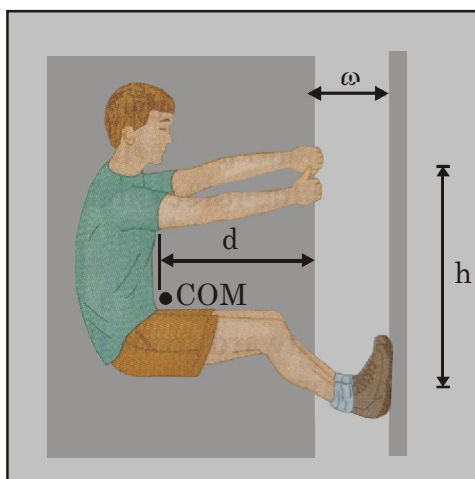
4. In nice weather, an electric field of magnitude 150 V/m is directed downwards near the Earth's surface. The field strength decreases uniformly with height and is 100 V/m at a height of 100 m. Assume earth to be flat.
- (A) The charge density in atmosphere is negative.
- (B) The charge density in atmosphere is same at all heights.
- (C) Charge density at height of 100 m is $\frac{\epsilon_0}{2}$.
- (D) A neutral metallic plate of thickness 1mm experiences a pressure of $0.05\epsilon_0$ when it is kept at a height of 10 m.
5. A thin ring of radius R has been uniformly charged with an amount of electric charge Q and placed near to a conducting sphere of radius R in such a way that the center of sphere O, lies on the axis at a distance of ℓ from the plane of the ring (figure)



- (A) The potential of sphere is $\frac{kQ}{\sqrt{\ell^2 + R^2}}$.
- (B) If the sphere is grounded, total charge on it is $-\frac{QR}{\sqrt{\ell^2 + R^2}}$.
- (C) The electric field at the centre of sphere due to induced charges is zero.
- (D) The electric field at P just outside sphere due to induced charges on sphere is $\frac{kQ(\ell - R)}{(R^2 + (\ell - R)^2)^{3/2}}$ towards left.

Space for Rough Work

6. In figure, a 56 kg rock climber is in a lie-back climb along a fissure, with hands pulling on one side of the fissure and feet pressed against the opposite side. The fissure has width $\omega = 0.20\text{m}$ and the center of mass of the climber is at a horizontal distance $d = 0.40\text{ m}$ from the fissure. The coefficient of static friction between hands and rock is $\mu_1 = 0.40$ and between boots and rock it is $\mu_2 = 1.2$. He is stable.



- (A) The least horizontal pull by hands and push by feet so that climber is stable is 350 N.
 (B) h must be more than or equal to 0.88 m.
 (C) h must be less than or equal to 0.88 m
 (D) The least horizontal pull by hands and push by feet so that climber is stable is 420 N.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

An isolated cube of non-conducting material was charged uniformly in volume. Corners of cube have the same potentials V . The electric field vectors near the corners of the cube (outside the volume of cube) have the same value E . Cube is now sawed into 8 equal parts in three mutually perpendicular flat sections (into 8 identical cubes) and one of the parts was removed.

7. What will be potential of the point which was the center of the former cube ?

(A) $\frac{V}{8}$ (B) $\frac{7V}{8}$ (C) $\frac{3V}{4}$ (D) $\frac{7V}{4}$

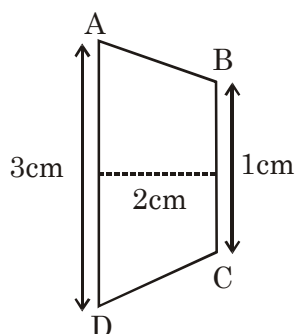
8. What will be the magnitude of the electric field vector at the same point ?

(A) $\frac{E}{4}$ (B) $\frac{7E}{8}$ (C) $\frac{E}{2}$ (D) $\frac{3E}{4}$

Space for Rough Work

Paragraph for Questions 9 and 10

A plane trapezium is in a plane containing principal axis of a lens. Its real image is formed. The image is of the shape of a square. Following statements are made about the lens and image. (Assume paraxial ray approximation to be always valid)



- (i) The lens must be converging.
- (ii) The lens must be kept to the right of trapezium
- (iii) BC must be perpendicular to the optic axis.
- (iv) AB & CD extended must meet at the object focus of the lens.

9. The correct statements are :-

- (A) All four (B) (i), (iv) (C) (i), (ii) (D) (i), (ii), (iv)

10. The side length of square will be :-

- (A) 1.5 cm (B) 2 cm (C) 3 cm (D) 6 cm

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

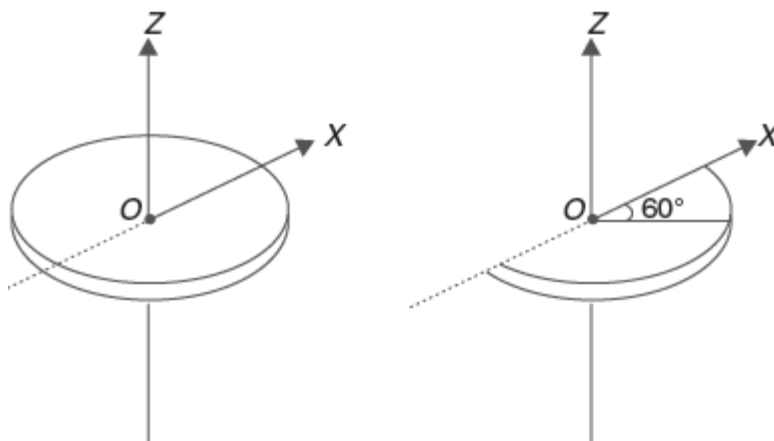
- The height of geostationary orbit above the surface of the earth is 36000 km. Radius of the earth is 6400 km. The earth shrinks to half its present radius (mass remaining unchanged).

If r (in km) is the radius of the geostationary satellite of the earth. Find $\frac{r^3}{1000}$.

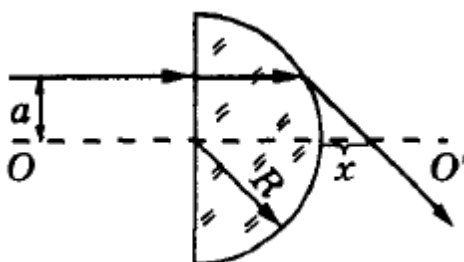
Space for Rough Work

2. An Indian bread "Roti" is a uniform disc of mass M and radius R . Before eating a person usually folds it about its diameter (say about x axis). After folding it a sector of angle 60° is removed from it. Find the moment of inertia of the remaining "Roti" about Z -axis. If answer is

$$\frac{MR^2}{n}, \text{ fill value of } n.$$



3. A Ray of light falls normally on a flat surface of a glass hemisphere of radius 4 cm. The distance from the point of incidence of the beam to the axis of the hemisphere $a = 2$ cm, index of refraction $n = \sqrt{2}$. At what distance x (in cm) from the curved surface of the hemisphere does the beam cross the axis ?



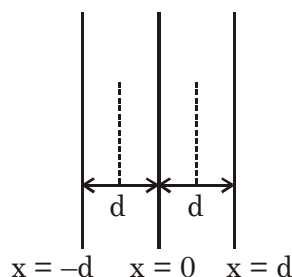
Space for Rough Work

4. A copper wire of length 5cm carries a current of density $j = 1 \text{ A (mm)}^{-2}$. The density and molar mass of copper are 9000 Kg m^{-3} and 63 gmol^{-1} . Each copper atom contributes one free electron. The temperature of the wire is 27°C . Estimate the (average) distance in (10^4 km) travelled by a free electron during the time it moves from one end of the copper wire to the other end. Assume that thermal motion of electrons are similar to that of molecules of an ideal gas.

Take molar mass of electron = $\frac{1}{500\pi} \text{ gm}$

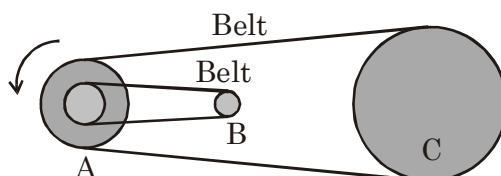
5. The space $0 < x < d$ has uniform charge density ρ ($\rho > 0$), and the space $-d < x < 0$ has uniform charge density $-\rho$. There is no charge in the regions $-\infty < x < -d$ and $d < x < \infty$. In the region $x > d$, an electron with mass m and charge $-e$ moves with its velocity vector oriented directly towards the charged space. What is the minimum initial velocity that the electron must have if it

is able to pass through the charged space ? If that velocity is $x d \sqrt{\frac{e\rho}{m \epsilon_0}}$ fill x.

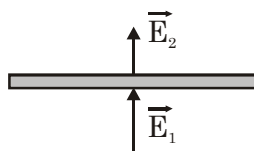


Space for Rough Work

6. Figure shows three rotating uniform disks that are coupled by belts. One belt turns around the rims of disks A and C. Another belt runs around a central hub on disk A and the rim of disk B. The belts move smoothly without slippage on the rims and hub. Disk A has radius R , its hub has radius $0.5000 R$; disk B has radius $0.2500 R$ and disk C has radius $2.000 R$. Disks B and C have the same density (mass per unit volume) and thickness. What is the ratio of the magnitude of the angular momentum of disk C to that of disk B? Hub is massless



7. A source of light is a thin thread of length $\ell = 10$ cm located on the principal axis of a converging lens with focal length $f = 5$ cm and diameter $D = 1$ cm. The near end of the thread is located a distance $a = 10$ cm from the lens. Find the minimum size (diameter in mm) of the illuminated spot on a screen placed on the other side of the lens perpendicular to the lens principal axis.
8. The electric field in space is produced by an external uniform electric field and the field of a charged metal plate (which can be considered uniform) as shown in figure. The electric field above the plate $E_2 = 2 \times 10^4$ V/m, and under it $E_1 = 5 \times 10^4$ V/m. What is the charge (in μC) on the plate, if the force exerted on the plate by the external field is 0.7 N.



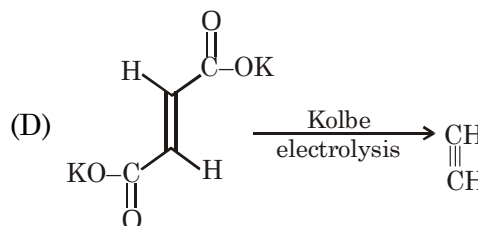
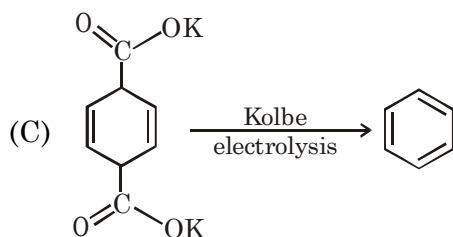
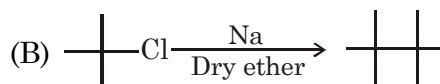
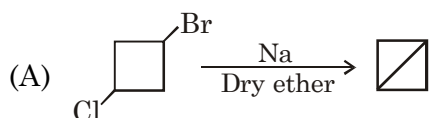
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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1. Which of following reaction(s) is/are correctly match with their major product :



Space for Rough Work

2. Choose the **CORRECT** statement among the following.

(A) Among $O = PBr_3$, $O = PCl_3$, $O = PF_3$

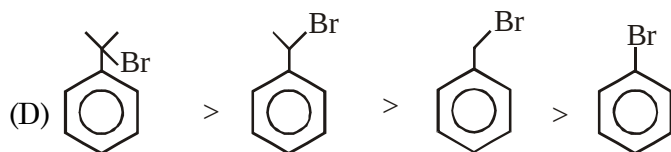
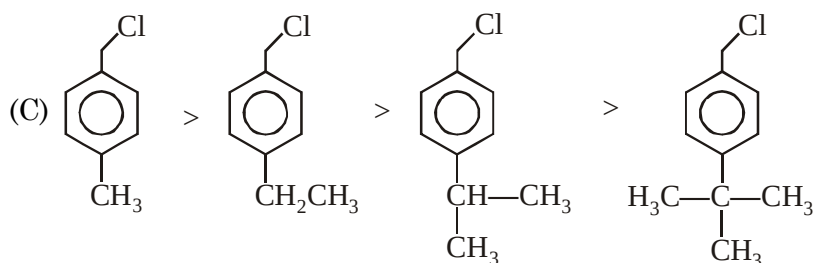
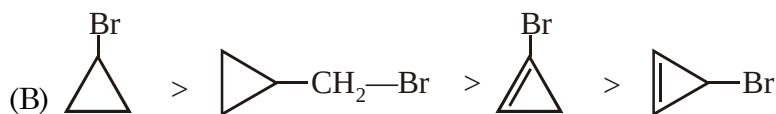
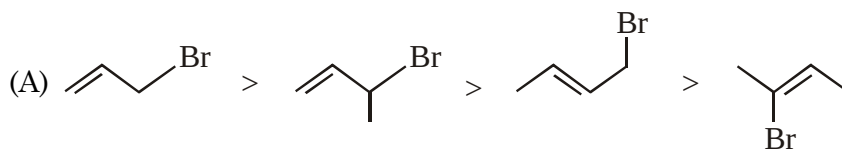
$P = O$ bond is fattest in $O = PF_3$

(B) $CH_3\hat{P}CH_3$ bond angles are equal in $(CH_3)_3P(CF_3)_2$ molecule

(C) $SiO_4^{4-} > PO_4^{3-} > SO_4^{2-} > ClO_4^-$ tendency of polymerisation

(D) Graphite shows an anisotropic electrical conductivity

3. Correct reactivity order of S_N1 is :



Space for Rough Work

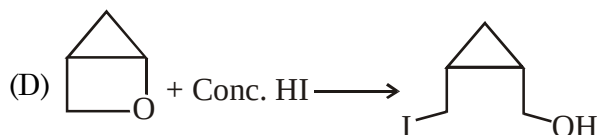
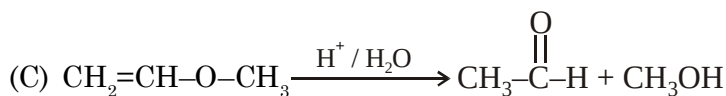
4. Total vapour pressure of binary solution of two volatile non electrolyte components A and B is given by :

$$P_T = 100 + 160 X_B$$

(X_B = mole fraction of B in liquid solution)

Correct option is/are -

- (A) B is more volatile than A
 (B) Vapour pressure of solution is more than pure A component
 (C) Vapour pressure of solution is more than pure B component
 (D) Vapour pressure pure A, $P_A^0 = 100$ unit while that of pure B, $P_B^0 = 260$ units.
5. Vapour pressure of pure liquids 'A' and 'B' are 300 torr and 800 torr respectively at 25°C. When these two liquids are mixed at this temperature to form a solution in which mole percentage of 'B' is 92, then the total vapour pressure is observed to be 0.95 atm. Which of the following is true for this solution,
- (A) $\Delta V_{\text{mix}} > 0$ (B) $\Delta H_{\text{mix}} < 0$ (C) $\Delta V_{\text{mix}} = 0$ (D) $\Delta H_{\text{mix}} = 0$
6. For the given reactions, correct major product is/are :

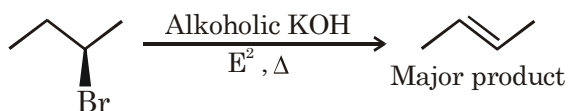


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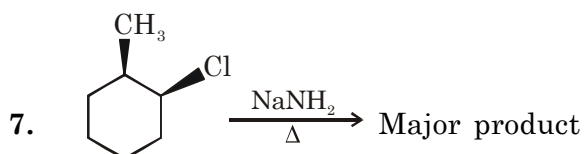
SECTION-I(ii) : (Maximum Marks : 12)

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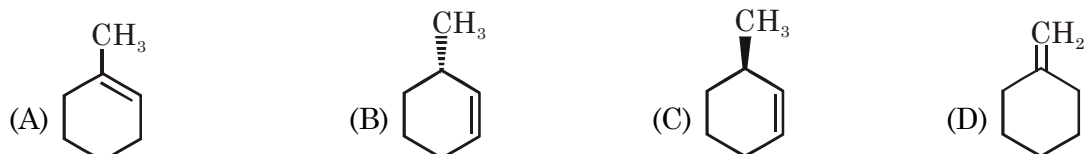
Paragraph for Questions 7 and 8



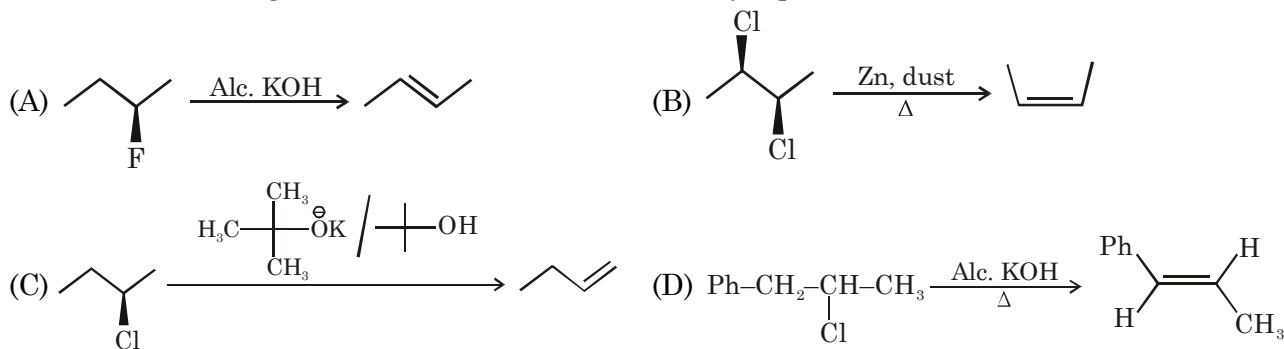
E² elimination is the example of anti elimination in which generally saytzeff's alkene as major product.



Major product is :



8. Match the following reaction with their incorrect major product :



Space for Rough Work

Paragraph for Questions 9 and 10

Valence bond theory is one of the two quantum mechanical approaches that explains bonding in molecules. But in some cases it is not adequate to explain the bonding parameters of molecules. Hybridisation of bonding orbitals also play a very important role in determining the value of bond angles & bond length.

9. Select the **CORRECT** statement.

(A) In $\text{BF}_2\text{-NH}_2$ molecule, FBF bond angle $< 120^\circ$

(B) $\text{NO}_2^- > \text{NO}_3^-$ (bond angle)

(C) $\text{H}_2\text{Se} > \text{PH}_3$ (bond angle)

(D) In OSF_4 , $\text{F} - \hat{\text{S}} - \text{F}$ equatorial bond angle is greater than 120°

10. In neutral molecule $\text{XeO}_{n_1}\text{F}_{n_2}$, central atom has no lone pair and ratio of $\frac{n_2}{n_1}$ is two, then which of the following orbital does not participate in hybridisation.

(n_1 & n_2 are non zero integers)

(A) p_z

(B) d_{z^2}

(C) $d_{x^2-y^2}$

(D) d_{xy}

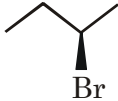
Space for Rough Work

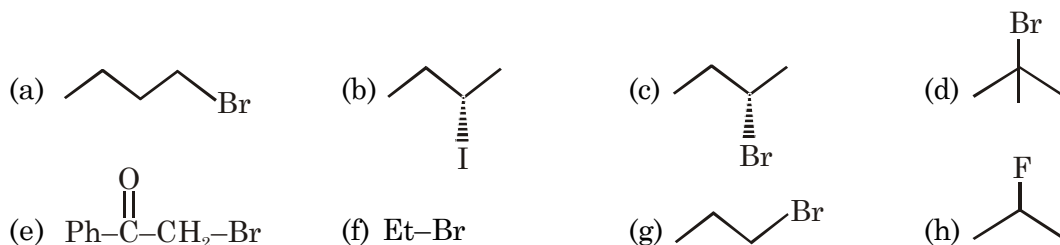
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS. **For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

+										-									
●	●	0	0	0	0	0	0	0	0	●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

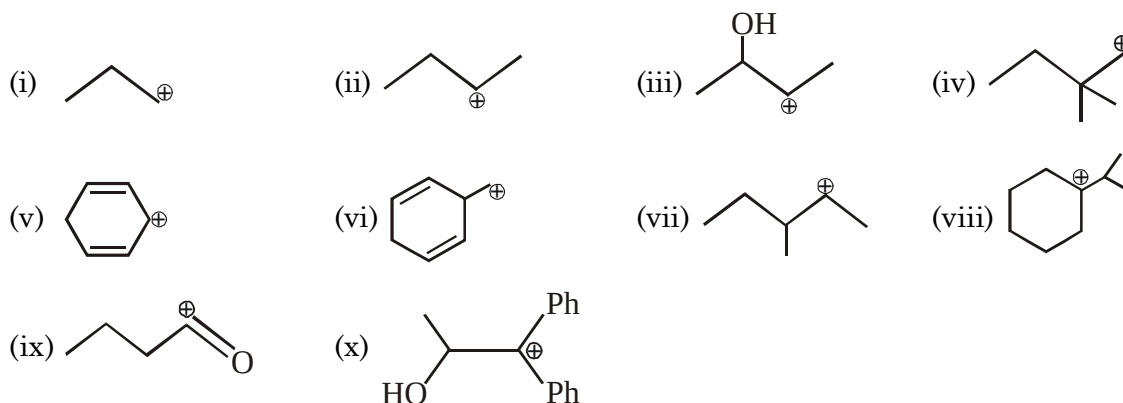
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If second nearest neighbour distance in a metallic fcc crystal is $200\sqrt{2}$ pm, find nearest distance between two octahedral voids in the metal, in Å ?
2. Count compounds from following list which give S_N2 reaction at faster rate than  :



Space for Rough Work

3. 0.001 mole of Potassium acid oxalate $K_2C_2O_4 \cdot 3H_2C_2O_4 \cdot 4H_2O$ can be oxidized by MnO_4^- in acid medium. Calculate the volume (in ml) of 0.2 M $KMnO_4$ needed in acid solution.
4. How many carbocation can undergo rearrangement :

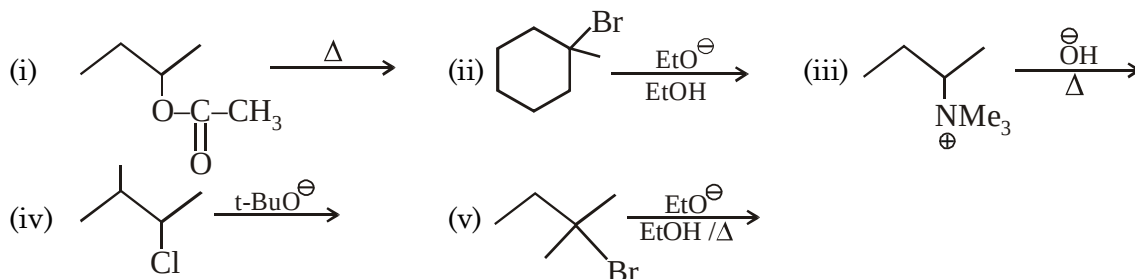


5. Find the number of **CORRECT** statements among the following.
- Generally, the atoms having high IE possess high EA but it is violated in the elements of group 2, group 15 and inert gases
 - Successive ionisation energies are always gradually becoming higher and higher for all elements
 - The second electron gain enthalpy even for non-metals is always negative
 - OF_2 is fluoride while OCF_2 is an oxide
 - The Z_{eff} of ns and (n-1)d electrons of Sc, Y and La are same

Space for Rough Work

6. Photons having energy equivalent to binding energy of 4th state He⁺ ion is used to irradiate a metal surface of Work function 1.4 eV. If electrons are further accelerated through a potential of 4 volts, then the minimum value of de broglie wavelength associated with the electron will be (in Å)

7. How many reaction(s) will give Hoffmann product as major alkene :



8. 75 ml, $\frac{M}{2}$ Ba(MnO₄)₂ solution in acidic medium can exactly neutralise how many of the following solution.

- (a) 150 ml of 1M FeSO₄ solution (b) 250 ml of 1M FeCrO₄ solution
(c) 187.5 ml of 1M K₂SO₃ solution (d) 375 ml of 1N K₂Cr₂O₇ solution

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If triangle ABC has sides $AB = 6$, $AC = 5$ and $BC = 4$. Points P_1 , P_2 and P_3 on BC satisfy $BP_1 = P_1P_2 = P_2P_3 = P_3C$. Then which of the following is/are true -

(A) $\cos B = \frac{9}{16}$

(B) $2(AP_1^2 + AP_2^2 + AP_3^2) = 163$

(C) $\cos B = \frac{1}{8}$

(D) $\cos C = \frac{1}{8}$

2. Which of the following cubic polynomials given below do not have their roots in arithmetic progression

(A) $x^3 + 3x^2 - 2x - 1$

(B) $x^3 + 3x^2 - 2x - 2$

(C) $x^3 + 3x^2 - 2x - 3$

(D) $x^3 + 3x^2 - 2x - 4$

Space for Rough Work

3. For the quadratic equation $x^2 + (2 - m)x - 4 = 0$ in x identify the correct statement(s)-
 (A) if $m < 5$ then exactly one root is smaller than -1
 (B) if $m < 5$ then both roots are smaller than -1
 (C) if $m > 5$ then both roots are greater than -1
 (D) roots are distinct real numbers for each $m \in \mathbb{R}$
4. The value of $\sum_{\alpha=1}^{\infty} \left(\sum_{r=0}^{\alpha} \left(\frac{{}^{\alpha}C_r}{5^{\alpha}} \right) \right)$ is less than (where $\alpha \in \mathbb{N}$) -
 (A) $\frac{2}{5}$ (B) $\frac{4}{5}$ (C) $\frac{5}{6}$ (D) $\frac{3}{4}$
5. Words are formed using all the letters of the word 'RAMPAUL'. Let p denotes number of words in which no two vowel are together and q denotes number of words in which no two consonant are together, then-
 (A) $p = q$ (B) $p = 10q$ (C) $q = 10p$ (D) $p + q = 800$
6. If $\sin^{-1} \left(\frac{\sqrt{x}}{2} \right) + \sin^{-1} \left(\sqrt{1 - \frac{x}{4}} \right) + \tan^{-1}(y) = \frac{2\pi}{3}$, then
 (A) Maximum value of $x^2 + y^2$ is $\frac{49}{3}$ (B) Maximum value of $x^2 + y^2$ is 4
 (C) Minimum value of $x^2 + y^2$ is $\frac{1}{3}$ (D) Minimum value of $x^2 + y^2$ is 3

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Let $f : A \rightarrow B$ be defined by $f(x) = 2x^2 - 6x + 5$ and A be the longest subset of $[-100, 100]$, so that $f(x)$ is invertible. Let $g(x)$ be inverse of $f(x)$.

On the basis of above information, answer the following questions :

7. $g(x)$ is the function defined by -
- (A) $\frac{3}{2} + \sqrt{\frac{x}{2} - \frac{1}{4}}, x \geq \frac{1}{2}$ and is increasing (B) $\frac{3}{2} - \sqrt{\frac{x}{2} - \frac{1}{4}}, x \geq \frac{1}{2}$ and is decreasing
- (C) $\frac{3}{2} + \sqrt{\frac{x}{2} - \frac{1}{4}}, x \geq \frac{1}{2}$ and is decreasing (D) $\frac{3}{2} - \sqrt{\frac{x}{2} - \frac{1}{4}}, x \geq \frac{1}{2}$ and is increasing
8. Number of distinct intersection points of $f(x)$ & $g(x)$ is -
- (A) 0 (B) 1 (C) 2 (D) 3

Space for Rough Work

Paragraph for Questions 9 and 10

Consider a triangle ABC.

Let $x + y - 3 = 0$, $x - 4y + 7 = 0$ and $2x - y = 0$ respectively the equations of the altitudes BE, CF and AD on the sides AC, AB and BC.

9. If one of the vertex of triangle is of the form $(\alpha, 2\alpha)$, then the locus of centroid of triangle ABC as α varies is -
- (A) $9x - 63y + 129 = 0$ (B) $9x + 63y - 129 = 0$
(C) $9x + 45y - 89 = 0$ (D) $9x + 45y - 99 = 0$
10. Let Q be any point on the line L : $x - k = 0$, where k is abscissa of orthocenter of $\triangle ABC$, P is a point $(k, 0)$ other than point Q, QR is the acute angle bisector of angle OQP, (O is origin) meets the x-axis at point R, then the locus of foot of perpendicular from R on the OQ is -
- (A) $(x^2 + y^2)(x + 2) + x = 0$ (B) $(x^2 + y^2)(x - 1) + x = 0$
(C) $(x^2 + y^2)(x - 2) + x = 0$ (D) $(x^2 + y^2)(x + 1) + x = 0$
-

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Suppose that a, b, c are positive integers such that $a(\log_{144} 3) + b(\log_{144} 2) = c$, then value of $\frac{a+b}{c}$ is
2. If x is an odd integer and y, z are even integers, satisfying $x + y + z = 47$ and $x, y, z \geq 0$, then number of solutions is
3. Let α, β are roots of $x^2 - 3x + 5 = 0$. If the value of $\frac{\alpha^6 \beta + \alpha \beta^6}{\alpha + \beta}$ is k , then the value of $(100 - |k|)$ is
4. A class has 4 teachers & 8 students. Number of ways in which they can be seated in a line, if in between any two consecutive teachers there are exactly two students is $k \cdot 9!$, then k is equal to

Space for Rough Work

5. One side of a triangle has length 4 units and remaining two sides are of length x and y units.
If perimeter of triangle is 18 and the area of triangle is maximum, then $\frac{x \cdot y}{7}$ is equal to
6. If lines $3x + 2y = 10$ and $-3x + 2y = 10$ are tangents at the extremities of latus rectum of an ellipse whose centre is origin, then the length of latus rectum of ellipse is
7. If equation of common tangent to parabola $y^2 - 8x = 0$ and hyperbola $y^2 - 3x^2 + 3 = 0$ is $2x + \frac{cy}{\sqrt{2}} + 1 = 0$ ($c \in \mathbb{R}^+$), then c is equal to
8. Let x_1, x_2, x_3 be the solution of $\tan^{-1}\left(\frac{2x+1}{x+1}\right) + \tan^{-1}\left(\frac{2x-1}{x-1}\right) = 2\tan^{-1}(x+1)$ where $x_1 < x_2 < x_3$
then $2x_1 + x_2 + x_3^2$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator